

NAME: MUHAMMAD USMAN

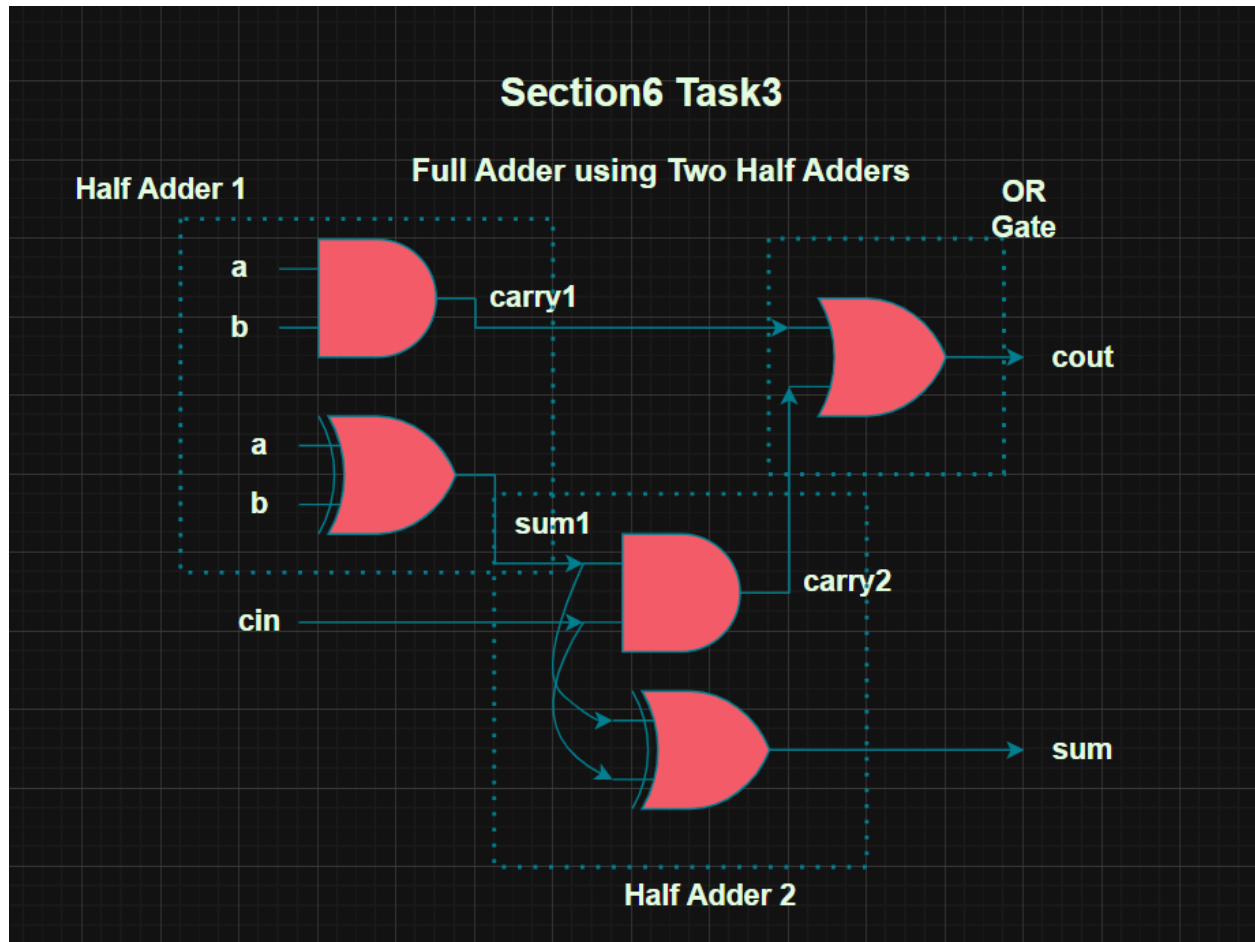
REG. NO# 2024-EE-171

1-EDA Playground link of Task3: <https://edaplayground.com/x/bhvi>

EDA Playground link of Task4: <https://edaplayground.com/x/fYsS>

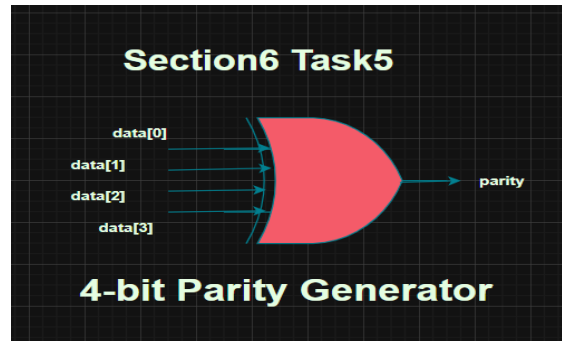
EDA Playground link of Task5: <https://edaplayground.com/x/KZda>

2-Schematic Diagram:



Task3_schematic_diagram.png

- Task4's Schematic Diagram not added because it's just conversion of equation, for making circuit so that it will only contain XOR-Gates.
- (Boolean Algebra / De Morgan Steps are provided within the Explanation Heading)



Task5_schematic_diagram.png

3-Output Screenshot:

```
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full164; Runtime version X-2025.06-SP1_Full164; Jul 17 10:47 2026
[PASS]: a=0 b=0 | assign(sum=0, carry=0) | proc(sum=0, carry=0)
[PASS]: a=0 b=1 | assign(sum=1, carry=0) | proc(sum=1, carry=0)
[PASS]: a=1 b=0 | assign(sum=1, carry=0) | proc(sum=1, carry=0)
[PASS]: a=1 b=1 | assign(sum=0, carry=1) | proc(sum=0, carry=1)
$stop at time 40 Scope: tb File: testbench.sv Line: 25
uccli% Finding VCD file...
./dump.vcd
[2026-07-17 14:47:59 UTC] Opening EPWave...
Done
```

Task3_ha_log_result.png

```
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full164; Runtime version X-2025.06-SP1_Full164; Jul 17 12:09 2026
[PASS] HA: a=0 b=0 | assign(sum=0, carry=0) | proc(sum=0, carry=0)
[PASS] HA: a=0 b=1 | assign(sum=1, carry=0) | proc(sum=1, carry=0)
[PASS] HA: a=1 b=0 | assign(sum=1, carry=0) | proc(sum=1, carry=0)
[PASS] HA: a=1 b=1 | assign(sum=0, carry=1) | proc(sum=0, carry=1)
[PASS] FA: a=0 b=0 cin=0 having sum=0 cout=0
[PASS] FA: a=0 b=0 cin=1 having sum=1 cout=0
[PASS] FA: a=0 b=1 cin=0 having sum=1 cout=0
[PASS] FA: a=0 b=1 cin=1 having sum=0 cout=1
[PASS] FA: a=1 b=0 cin=0 having sum=1 cout=0
[PASS] FA: a=1 b=0 cin=1 having sum=0 cout=1
[PASS] FA: a=1 b=1 cin=0 having sum=0 cout=1
[PASS] FA: a=1 b=1 cin=1 having sum=1 cout=1
$stop at time 120 Scope: tb File: testbench.sv Line: 46
uccli% Finding VCD file...
./dump.vcd
[2026-07-17 16:09:42 UTC] Opening EPWave...
Done
```

Task3_fa_log_result.png

LogShare

Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64; Jul 17 11:29 2026
[PASS]: a=0 b=0 c=0 d=0 | direct=0 | nor=0
[PASS]: a=0 b=0 c=0 d=1 | direct=1 | nor=1
[PASS]: a=0 b=0 c=1 d=0 | direct=1 | nor=1
[PASS]: a=0 b=0 c=1 d=1 | direct=1 | nor=1
[PASS]: a=0 b=1 c=0 d=0 | direct=1 | nor=1
[PASS]: a=0 b=1 c=0 d=1 | direct=1 | nor=1
[PASS]: a=0 b=1 c=1 d=0 | direct=1 | nor=1
[PASS]: a=0 b=1 c=1 d=1 | direct=1 | nor=1
[PASS]: a=1 b=0 c=0 d=0 | direct=0 | nor=0
[PASS]: a=1 b=0 c=0 d=1 | direct=0 | nor=0
[PASS]: a=1 b=0 c=1 d=0 | direct=0 | nor=0
[PASS]: a=1 b=0 c=1 d=1 | direct=1 | nor=1
[PASS]: a=1 b=1 c=0 d=0 | direct=0 | nor=0
[PASS]: a=1 b=1 c=0 d=1 | direct=1 | nor=1
[PASS]: a=1 b=1 c=1 d=0 | direct=0 | nor=0
[PASS]: a=1 b=1 c=1 d=1 | direct=1 | nor=1
\$stop at time 160 Scope: tb File: testbench.sv Line: 21
ucli% Finding VCD file...
./dump.vcd
[2026-07-17 15:29:18 UTC] Opening EPWave...
Done

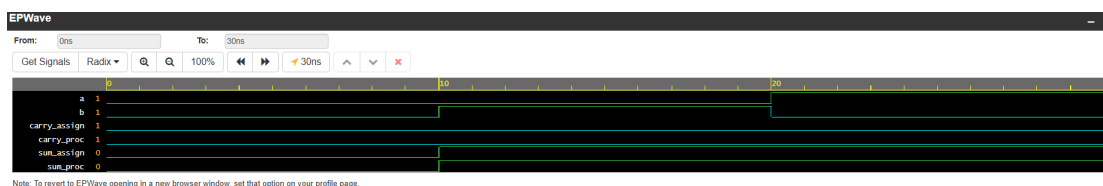
Task4_log_result.png

LogShare

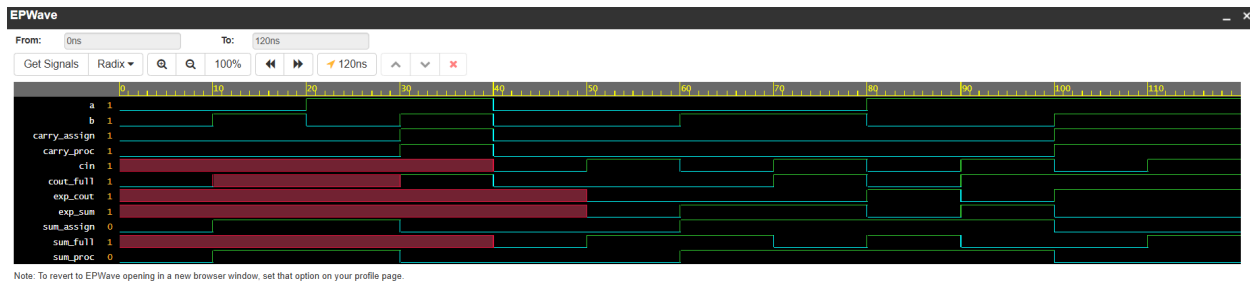
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64; Jul 17 11:46 2026
[PASS] data=0 (0) having parity=0
[PASS] data=1 (1) having parity=1
[PASS] data=10 (2) having parity=1
[PASS] data=11 (3) having parity=0
[PASS] data=100 (4) having parity=1
[PASS] data=101 (5) having parity=0
[PASS] data=110 (6) having parity=0
[PASS] data=111 (7) having parity=1
[PASS] data=1000 (8) having parity=1
[PASS] data=1001 (9) having parity=0
[PASS] data=1010 (10) having parity=0
[PASS] data=1011 (11) having parity=1
[PASS] data=1100 (12) having parity=0
[PASS] data=1101 (13) having parity=1
[PASS] data=1110 (14) having parity=1
[PASS] data=1111 (15) having parity=0
\$stop at time 170 Scope: tb File: testbench.sv Line: 28
ucli% Finding VCD file...
./dump.vcd
[2026-07-17 15:46:20 UTC] Opening EPWave...
Done

Task5_log_result.png

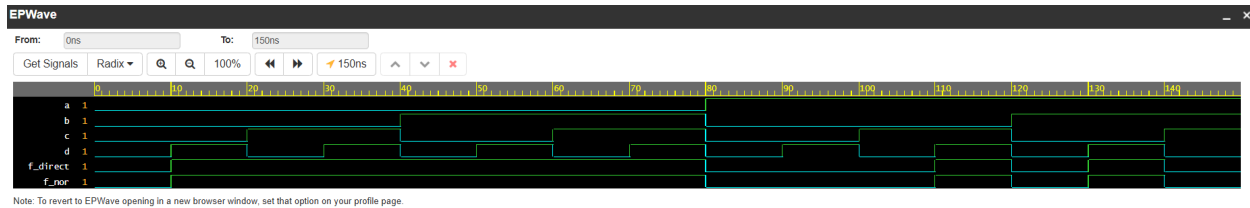
4-Waveform Screenshot:



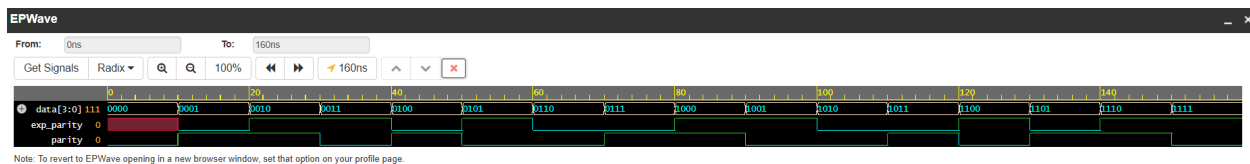
Task3_ha_waveform.png



Task3_fa_waveform.png



Task4_waveform.png



Task5_waveform.png

5-Explanation:

i-Task3 Reasoning:

In my point of view, the assign version and the always_comb version synthesize to the same gates even though they are written differently because when we synthesize our design in any software it will automatically synthesize our design in such a way that it will be the optimized version of our design, so, it will synthesize to the same gates as the original idea is same.

ii-Task4 Boolean Algebra / De Morgan Steps:

We start with:

$$f = (b + c + d) \cdot (a' + b + c) \cdot (a' + d)$$

- Step 1 – Identify the three OR terms (literals):

Let:

$$X = (b + c + d)$$

$$Y = (a' + b + c)$$

$$Z = (a' + d)$$

Now $f = X \cdot Y \cdot Z$.

- **Step 2 – Apply De Morgan's theorem to the whole AND:**

De Morgan says: $X \cdot Y \cdot Z = (X' + Y' + Z')'$

So:

$$f = ((b+c+d)' + (a'+b+c)' + (a'+d)')'$$

- **Step 3 – Interpret the terms as NOR outputs:**

- $(b+c+d)'$ is a 3-input NOR of b, c, d

- $(a'+b+c)'$ is a 3-input NOR of a', b, c

- $(a'+d)'$ is a 2-input NOR of a' and d

(But we need a' first. We can get a' using a NOR gate with both inputs tied to a, i.e. $a' = (a+a)' = \text{NOR}(a,a)$)

- **Step 4 – Final stage:**

The outer complement $(\dots)'$ is a 3-input NOR that combines the three intermediate results.

Resulting all-NOR network:

$$a_n = \text{NOR}(a, a) \rightarrow \text{this gives } a'$$

$$n1 = \text{NOR}(b, c, d) \rightarrow \text{gives } (b+c+d)'$$

$$n2 = \text{NOR}(a_n, b, c) \rightarrow \text{gives } (a'+b+c)'$$

$$n3 = \text{NOR}(a_n, d) \rightarrow \text{gives } (a'+d)'$$

$$f = \text{NOR}(n1, n2, n3) \rightarrow \text{final output}$$